

## High prevalence of oncogenic HPV-16 in cervical smears of asymptomatic women of eastern Uttar Pradesh, India: a population-based study.



In developing countries like India, occurrence of Human papillomavirus (HPV) in cervical cancer as well as in the asymptomatic population was observed to be very high. Studies on HPV prevalence have been conducted in different parts of the country but no data were available from the eastern region of Uttar Pradesh (UP). The present study aimed to determine the status of HPV prevalence and its association with different socio-demographic factors in this population. Prevalence of HPV was investigated in a total of 2424 cervical scrape samples of asymptomatic women. Primer sets from L1 consensus region of viral genome were used to detect the presence of HPV, and the positive samples were genotyped by sequencing. Univariate binary logistic regression analysis was used to evaluate association of socio-demographic factors with HPV. 9.9% of the clinically asymptomatic women were found to be infected with HPV comprising 26 different genotypes. Among HPV-positive women, 80.8% showed single infection, while 15.4% harboured multiple infections. HPV-16 (63.7%) was the most prevalent, followed by HPV-31 (6.7%), HPV-6 (5.4%), HPV-81 (4.6%) and HPV-33 (4.2%). Significant association of HPV with non-vegetarian diet ( $P$  less than 0.05) and rural residential areas ( $P$  less than 0.01) were observed. High prevalence of HPV-16 in asymptomatic women of this population, a frequency comparable to invasive cervical cancers, highlights an urgent need for a therapeutic HPV vaccine covering HPV-16 and other high-risk types to provide protection against the disease.